

Analyzing Global News Discourse on the Russo-Ukrainian War (2022–2025): A Keyword, Topic, and Sentiment Exploration

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Abstract—This study analyzed global news coverage of the Russo-Ukrainian War from 2022-2025. Researchers employed custom Python code to extract war-related news headlines from three major news networks: BBC, The Guardian, and The New York Times. Through natural language processing techniques, we classified the headlines into five major topics, tracked sentiment changes over time, and identified the countries, leaders, and Ukrainian regions that received the most media exposure. This research provides insights into how major mainstream media outlets report on international political events.

Index Terms—Global Analysis, Sentimental Analysis, Topic Modeling, Natural Language Processing, Latent Dirichlet Allocation

The code can be found in this link and available at https://github.com/zhian-li0126/STA220_Project.git

I. INTRODUCTION

OVER the past several years, the Russo-Ukrainian War has drawn considerable global attention, shaping international relations and dominating the news cycle in major media outlets. From military developments on the ground to diplomatic negotiations and humanitarian repercussions, the conflict has been portrayed through countless news articles and editorials. During the period from 2022 to 2025, three prominent news sources—BBC, The Guardian, and The New York Times—have served as primary conduits for information and analysis on the evolving situation. Given their international readership and influence, these outlets play a pivotal role in shaping public perception and policy discussions.

In this report, we undertake a comprehensive examination of the global news discourse surrounding the Russo-Ukrainian War by collecting and analyzing articles across the aforementioned media platforms. Through keyword extraction, sentiment analysis, topic modeling, and discussions of public attitudes, this study aims to capture the key themes and sentiments propagated in world media. The overarching objective is to highlight how the coverage, language, and framing of events contribute to international conversations on the conflict, offering insights into patterns of reporting and public engagement.

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II. MATERIALS AND METHODS

A. Scope and Data Collection

The scope of this study covers media coverage on the Russo-Ukrainian War from 2022 to 2025, focusing on three prominent international news outlets: BBC, The Guardian, and The New York Times. By spanning multiple years, the collected dataset captures evolving narratives and sentiments related to key events, negotiations, and public reactions over time.

Data collection involved a two-part team effort. One researcher was responsible for scraping and curating articles from the BBC using BeautifulSoup, while another gathered articles from The Guardian and The New York Times using their documented APIs. Automated scripts were employed to systematically collect articles containing relevant keywords (e.g., “Ukraine,” “Russia,” “conflict,” “war,” and related terms). Publication dates, titles, links, and any immediately identifiable keywords provided by the news sources were recorded for each article. Special care was taken to handle duplicate or near-duplicate items by comparing article titles and URLs, ensuring that only unique content remained.

All retrieved data were stored in a uniform JSON format:

```
{
  "Article Title 1": [
    "MM-DD-YYYY",
    "Identified Keywords (if applicable)",
    "URL",
    "Source/Journal"
  ],
  "Article Title 2": [
    "MM-DD-YYYY",
    "Identified Keywords (if applicable)",
    "URL",
    "Source/Journal"
  ],
  ...
}
```

This structure preserves important metadata for subsequent processing and analysis, enabling easy filtering and sorting by date, keyword, or publication source. As part of an initial

cleaning phase, the raw text of each article was also extracted (where feasible) for advanced methods such as topic modeling and sentiment analysis.

B. Analytical and Processing Approach

Upon completing the data collection, the text corpus was subjected to a series of preprocessing steps to enhance the reliability and relevance of subsequent analyses. Common natural language processing (NLP) techniques were applied—removing stop words, normalizing text via lowercasing, and performing tokenization—to reduce noise and facilitate comparability across articles. Where necessary, domain-specific stop words (e.g., “article,” “guardian,” “briefing,” etc.) were also excluded to ensure that repeated institutional references did not skew the results.

Advanced steps, such as named entity recognition (NER) and part-of-speech tagging, were performed to uncover entities (persons, locations, organizations) and key grammatical structures. By doing so, the study aimed to capture patterns not only in raw text frequency but also in how different subjects and entities are linked within the news coverage. The main analytical procedures are described in detail below.

1) Title Frequency Analysis:

To analyze key terms in article titles, we implemented a specialized text processing pipeline using NLTK and custom regex patterns. The process began with lowercase conversion followed by a two-stage name unification system that standardized references to political figures (e.g., converting both “Vladimir Putin” and “Putin” to “vladimir_putin”) and country abbreviations (e.g., “UK” to “united_kingdom”) to ensure consistent entity identification regardless of reference format and avoid redundancy and ambiguity in further interpretations.

The preprocessing pipeline included several noise reduction techniques. We removed standard English stopwords, filtered out verbs through part-of-speech tagging, and eliminated a custom list of low-information words commonly found in headlines (e.g., “new,” “day”) and publication-specific terms (e.g., “guardian,” “bbc”). After tokenization, we performed frequency counts on the remaining terms to identify the most common political actors, locations, and concepts appearing in headlines across the dataset. This quantitative approach provided an objective measure of the most prominently featured elements in coverage across the studied period and publications.

2) Latent Dirichlet Allocation (LDA) and Topic Modeling:

Latent Dirichlet Allocation (LDA) [1] was employed as a probabilistic modeling technique to uncover latent topics in the corpus. The implementation leveraged scikit-learn’s LDA module with a specialized processing pipeline tailored to our pre-tokenized data. Rather than applying standard tokenization within the model, we utilized a dummy tokenizer that preserved our carefully preprocessed tokens—which had already undergone name unification, stopword removal, and part-of-speech filtering. The document-term matrix was constructed

using CountVectorizer with custom parameters to accept pre-processed token lists directly. We empirically selected a model with five topic components ($n_components=5$) after testing various configurations to balance interpretability with thematic breadth. This approach allowed us to identify underlying themes—such as humanitarian issues, diplomatic dialogues, or military movements—without imposed categorization, letting the statistical patterns in the data reveal naturally occurring topic clusters. Each article was subsequently represented as a probability distribution across these topics, enabling quantitative comparison of thematic emphasis across different news sources and time periods [2].

Beyond basic keyword analysis and LDA, additional topic modeling techniques (e.g., non-negative matrix factorization) were tested to cross-validate results and ensure robustness of the findings. These methods facilitate a more nuanced understanding of how discussions evolve over time, highlighting new developments (e.g., peace negotiations, changing alliances, or public sentiment) as they appear in the news cycle. Topic modeling outputs were subsequently visualized in order to better communicate the distribution and evolution of topics. By juxtaposing topics across different news outlets, we can pinpoint potential divergences in editorial choices or regional emphases, thus painting a more holistic picture of global media discourse on the Russo-Ukrainian War.

3) Geospatial Analysis:

For visualizing our analytical findings, we employed a multi-tiered approach using Python’s data visualization libraries (primarily Matplotlib, Seaborn, and Plotly). Frequency distributions of entities were represented through horizontal bar charts to clearly illustrate the relative prominence of countries, political figures, and locations in the news coverage. We standardized the color schemes and formatting across visualizations to maintain consistency and facilitate comparison between different entity types. For geographical data, we implemented two complementary visualization strategies: a static heat map focused on Ukraine’s regional coverage using coordinates extracted from our processed dataset, and an interactive world map developed with Plotly and hosted on GitHub Pages that displays country-level mention frequencies on a logarithmic scale. The interactive visualization incorporates hover functionality that displays precise frequency counts and allows users to zoom into specific regions of interest.

III. RESULTS AND DISCUSSION

Our analysis encompassed 43,338 headlines related to the Russo-Ukrainian War published between 2022 and 2025, comprising 27,749 from The Guardian, 8,388 from BBC, and 7,201 from The New York Times, shown in Figure 1. This substantial dataset allowed us to identify persistent patterns in media coverage across multiple dimensions, including geopolitical framing, key actors, and thematic priorities.

A. Title Frequency Analysis

Analysis of term frequency in news headlines revealed clear patterns in media framing of the Russo-Ukrainian conflict. As

TABLE I
TOPIC MODELING RESULTS SHOWING THE TOP 10 KEYWORDS FOR EACH TOPIC WITH THEIR FREQUENCY COUNTS

Topic	Keywords (Frequency)
Topic 0	united_kingdom(562), prime_minister(484), donald_trump(452), european_union(448), global(417), china(389), oil(346), gas(337), nato(293), keir_starmer(281)
Topic 1	ukraine(6378), war(3524), russia(2601), vladimir_putin(1568), russian(1091), joe_biden(507), united_states(470), kyiv(464), military(444), zelenskiy(389)
Topic 2	war(830), donald_trump(803), israel(801), gaza(718), joe_biden(715), united_states(536), election(387), united_kingdom(332), hamas(327), bank(250)
Topic 3	ukraine(3207), russia(2429), russian(2272), war(2102), ukrainian(1611), invasion(827), nuclear(630), vladimir_putin(599), united_kingdom(581), united_states(521)
Topic 4	united_kingdom(764), energy(706), years(395), prices(308), wars(300), crisis(268), price(220), food(204), united_states(204), elizabeth_truss(202)

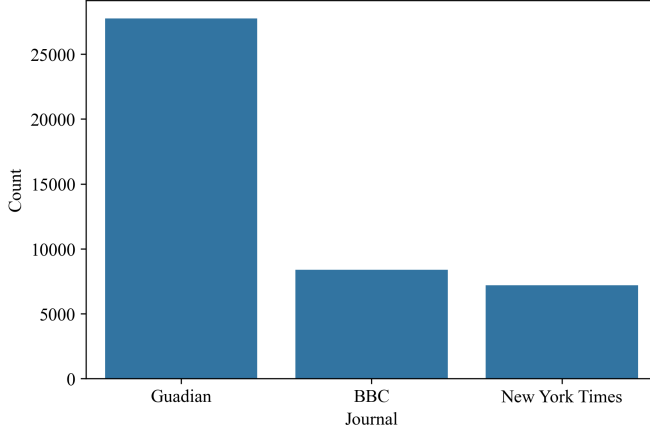


Fig. 1. Top words list mentioned in Russo-Ukraine war news titles

shown in Figure 2, "Ukraine" dominated coverage (9,586 occurrences), substantially outpacing "Russia" (5,200) and "war" (6,496), suggesting a media narrative centered on Ukraine's perspective rather than Russia's, as the war, portrayed as "invasion" (1,013) and "attack" (664) from Russia, corresponds with the center of focus "Ukraine" (9,586). Key international players featured prominently, with the "United Kingdom" (2,439) and "United States" (1,904) appearing frequently, reflecting their substantial military and diplomatic support. Among political leaders, "Vladimir Putin" received the most mentions (2,326), followed by "Donald Trump" (1,603) and "Joe Biden" (1,342), highlighting both the conflict's origins and the significance of American policy positions in the international response. The appearance of "Israel" (996) and "Gaza" (945) among the top terms reveals how media coverage frequently connected the Ukraine conflict with Middle Eastern tensions. Meanwhile, terms related to humanitarian and security concerns—"invasion" (1,013), "energy" (784), "crisis" (764), "attack" (664), "aid" (636), and "nuclear" (630)—underscore how headlines emphasized both immediate security threats and broader implications for regional stability and energy security. This frequency distribution demonstrates that news coverage framed the conflict as a full-scale war with significant global ramifications rather than a localized

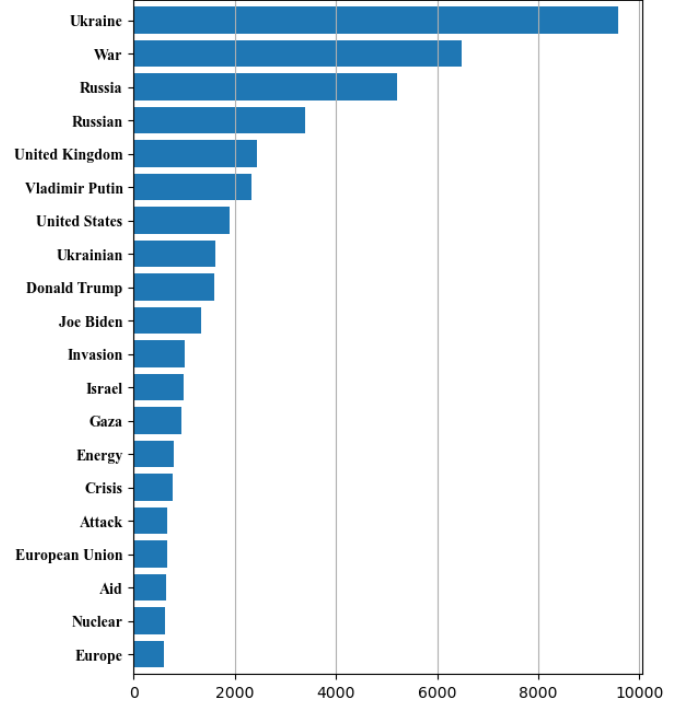


Fig. 2. Top words list mentioned in Russo-Ukraine war news titles

or limited engagement.

B. Topic Modeling

Latent Dirichlet Allocation (LDA) analysis revealed five distinct thematic clusters in Table I of our dataset, providing insight into how media coverage organized around key narratives. As shown in Figure 3, these topics represent coherent frames through which the conflict was presented to international audiences. Topic 1 emerged as the core war narrative, dominated by direct references to the primary actors ("ukraine," "russia," "vladimir_putin") and military aspects of the conflict. With "ukraine" appearing 6,378 times and "war" 3,524 times within this topic, it represents the central framing of the conflict as a direct military confrontation between the two nations, with significant attention to leadership figures including Zelenskiy and Biden in their roles as wartime

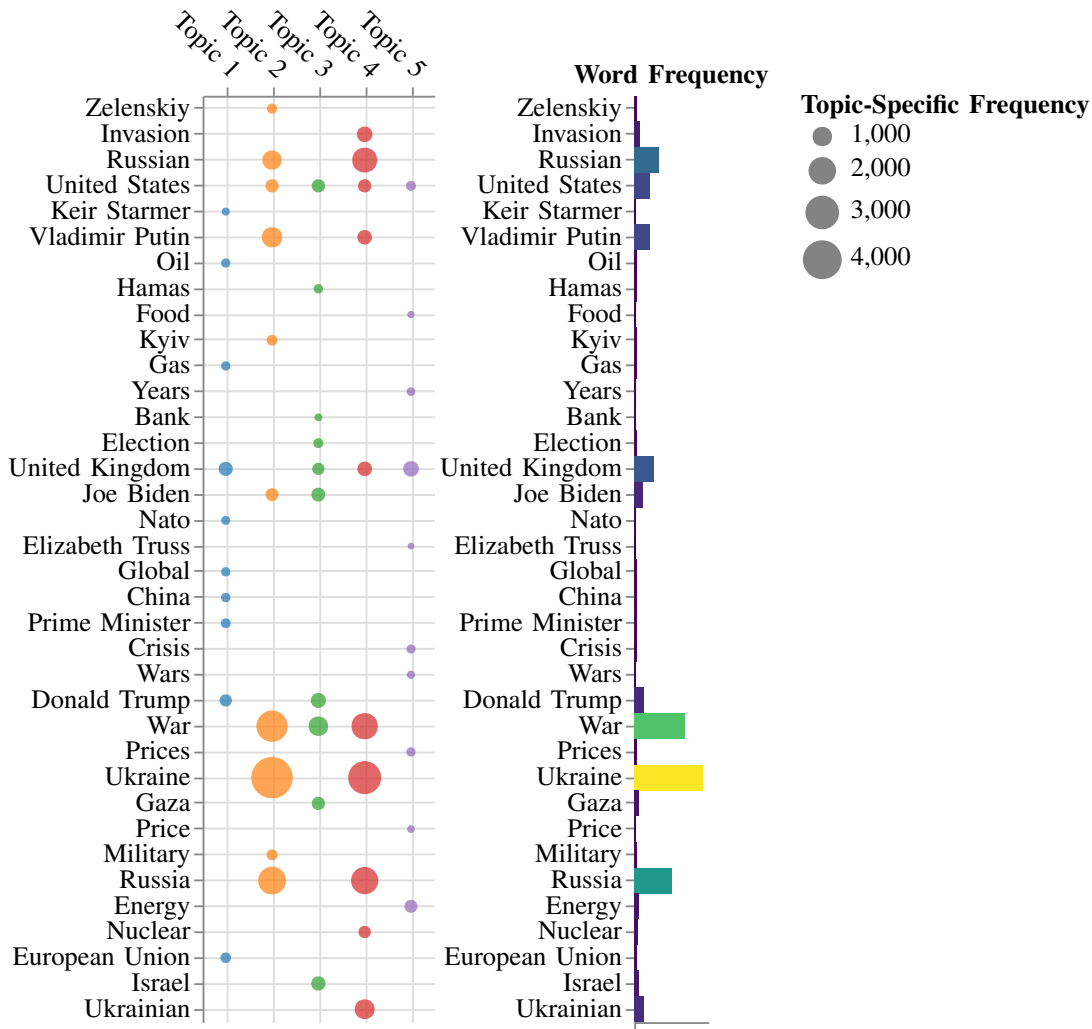


Fig. 3. Topic-Term Distribution Matrix (left) and Term Frequency (right) in the Topic LDA Model

leaders. Topic 3 similarly focused on the conflict itself but emphasized the invasion framing, with terms like "invasion" (827) and "nuclear" (630) suggesting coverage of both the initial aggression and subsequent security threats. The presence of "ukrainian" (1,611) alongside "russian" (2,272) indicates this topic captured coverage of both nations' perspectives and actions.

Topic 0 reflected the international diplomatic dimension, with strong representation of "united_kingdom" (562), "european_union" (448), and NATO (293), alongside energy security concerns through terms like "oil" (346) and "gas" (337). The prominence of "prime_minister" (484) and British political figures like "keir_starmer" (281) indicates significant coverage of the UK's diplomatic role and internal politics related to the conflict.

Topic 4 captured the economic impact narrative, with "energy" (706), "prices" (308), "crisis" (268), "price" (220), and "food" (204) highlighting how media coverage connected the war to broader economic consequences for civilians globally.

The presence of "elizabeth_truss" (202) likely reflects coverage during her brief tenure as UK Prime Minister when energy crisis management was paramount.

Topic 2 revealed an unexpected connection between the Ukraine conflict and the Israel-Gaza situation, with "israel" (801), "gaza" (718), and "hamas" (327) appearing prominently alongside "war" (830) and "donald_trump" (803). This suggests media coverage frequently drew parallels between these conflicts or discussed them within a common geopolitical framework, particularly in relation to US politics and Trump's positions on both conflicts.

The LDA results demonstrate how media coverage extended well beyond direct military reporting to encompass diplomatic, economic, and comparative conflict narratives, illustrating the multidimensional framing of the Russo-Ukrainian War in international media.

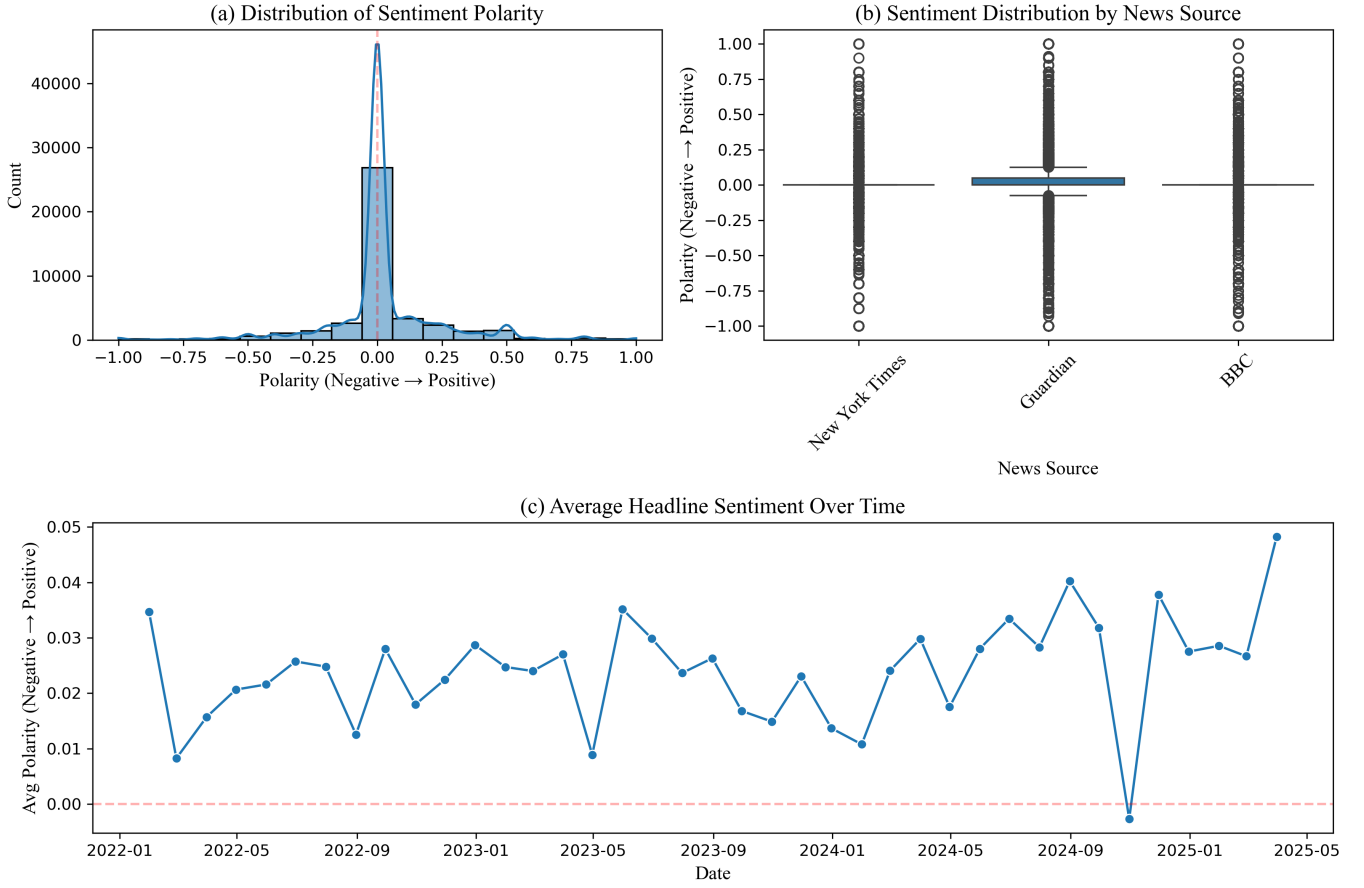


Fig. 4. Sentiment Analysis of News Headlines: (a) Sentiment Polarity Distribution, (b) Source-wise Comparison, and (c) Temporal Trends

C. Sentiment Analysis

To assess the emotional tone of media coverage, we conducted sentiment analysis on 42,726 headlines spanning from January 2022 to March 2025 across The New York Times, Guardian, and BBC. Using TextBlob’s natural language processing capabilities, we calculated polarity scores ranging from -1 (highly negative) to +1 (highly positive) for each headline. As shown in Figure 4, the distribution of sentiment polarity exhibited a pronounced concentration around neutral values, with 26,760 headlines (62.6%) classified as neutral, 9,473 (22.2%) as positive, and 6,493 (15.2%) as negative. This distribution suggests that while news outlets generally maintained neutrality in their headline framing, there was a slight tendency toward positive sentiment when non-neutral language was employed. Comparative analysis between sources revealed statistically significant differences in sentiment (ANOVA: $F=15.86$, $p<0.0001$), with The Guardian showing the highest average positive sentiment (0.027), followed by BBC (0.019) and The New York Times (0.011). The temporal trend analysis in Figure 4(c) demonstrated consistent positive sentiment across the studied period, with minor fluctuations corresponding to key events. Most notably, October 2024 showed the only instance

of negative average sentiment, potentially related to election-period escalations, before rebounding to the highest positive levels in February 2025. Tables II and III illustrate the top 5 emotionally charged headlines, with positive headlines often featuring themes of international solidarity and diplomatic progress, while negative headlines predominantly depicted humanitarian suffering and military violence, particularly in the conflict’s early phases.

D. Visualization and interpretation

1) *Top Countries or Regions Mentioned on Ukraine War Media Headline Coverage:* As one of the most significant armed conflicts of the 21st century, the Ukraine War has drawn involvement from numerous nations that have adopted various positions in the conflict. Our analysis of countries and regions, shown in Figure 5 most frequently referenced in Ukraine War media headlines reveals expected patterns alongside several noteworthy observations.

Unsurprisingly, Ukraine and Russia dominated the headline coverage as the primary belligerents. Beyond these direct participants, the United Kingdom and the United States featured prominently in media reporting, reflecting their substantial role in providing military assistance and maintaining support for

TABLE II
MOST POSITIVE HEADLINES

Title	Journal	Date	Sentiment_Polarity
Once Best Friends, Bulgaria Takes a Stand Against Russia	New York Times	2022-05-11	1.0
What's the Best Way to Treat Addiction?	New York Times	2022-05-12	1.0
How Ukraine's Greatest Novelist Is Fighting for His Country	New York Times	2022-05-24	1.0
'Safety would be the perfect Mother's Day gift'	BBC	2022-05-08	1.0
'He did his best': Macron's meeting with Trump prompts cautious optimism in France on Ukraine	Guardian	2025-02-25	1.0

TABLE III
MOST NEGATIVE HEADLINES

Title	Journal	Date	Sentiment_Polarity
I'm in Kyiv, and It Is Terrifying	New York Times	2022-02-25	-1.0
Videos show devastating strike at Mariupol hospital maternity ward.	New York Times	2022-03-09	-1.0
Its offensive slowed, Russia uses long-range missiles to devastating effect.	New York Times	2022-03-20	-1.0
'Evil Came': The Toll of Russia's Occupation in Kyiv's Suburbs	New York Times	2022-04-07	-1.0
Bernie Sanders dismisses Republicans' 'horrific' calls for Zelenskyy to resign	Guardian	2025-03-02	-1.0

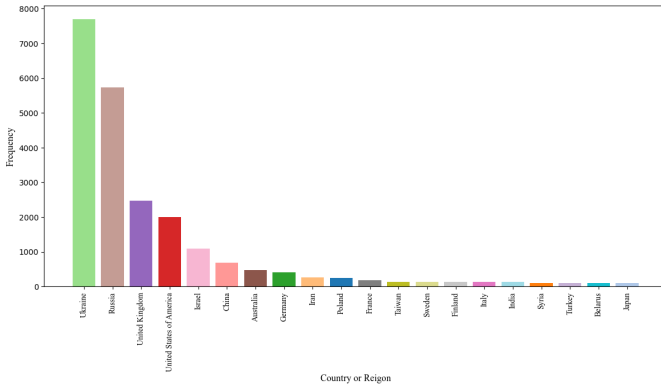


Fig. 5. Top countries list mentioned in Russo-Ukraine war news titles

Ukraine since the conflict's inception (particularly the United States prior to Donald Trump's administration).

Israel received considerable media attention, largely attributable to ongoing journalistic comparisons between the Israel-Gaza conflict and the Russia-Ukraine war. China's significant presence in headlines stems from its diplomatically complex position, with the international community closely monitoring its stance as both Ukraine and Russia sought to secure Chinese support, or at minimum, prevent alignment with their adversary.

Additional NATO member states and key Russian allies are also featured regularly in headlines. Notably, Syria and Taiwan—regions experiencing civil conflict and geopolitical tension respectively—appeared with unexpected frequency in the coverage. This suggests media organizations maintained awareness of other global flashpoints while reporting on developments in Ukraine, potentially drawing strategic parallels or identifying interconnected security implications across these distinct regional contexts.

2) *Top World Leaders Mentioned on Ukraine War Media Headline Coverage:* The frequency analysis of individuals

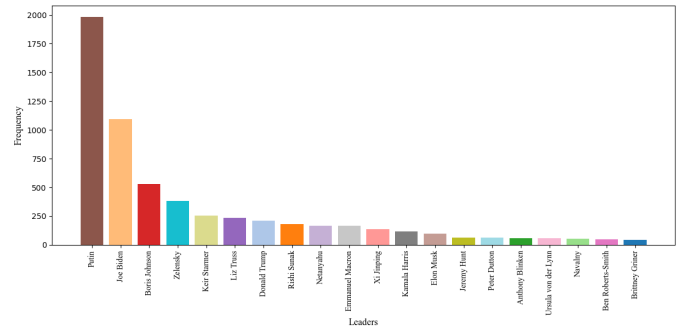


Fig. 6. Top world leaders list mentioned in Russo-Ukraine war news titles

mentioned in Ukraine war-related news headlines, shown in Figure 6 reveals a concentration on influential political leaders directly involved in the conflict and its diplomatic dimensions.

Vladimir Putin and Joe Biden predictably dominated the coverage, with Putin's role as the initiator of the invasion and Biden's position as commander-in-chief of the world's preeminent military power providing extensive military aid to Ukraine. President Volodymyr Zelenskyy featured prominently as Ukraine's wartime leader, instrumental in directing his nation's defensive campaign against Russian aggression.

British political figures—including Boris Johnson, Keir Starmer, Liz Truss, Rishi Sunak, and Jeremy Hunt—appeared with notable frequency, reflecting both the United Kingdom's substantial involvement in supporting Ukraine and the inherent bias in our data sources (BBC and The Guardian). This media-origin effect must be considered when interpreting the prominence of British politicians in the analysis.

Donald Trump's inclusion stems from his recent controversial exchanges with Zelenskyy, which generated significant international attention. Similarly, Benjamin Netanyahu, Emmanuel Macron, and Xi Jinping appeared regularly in headlines, consistent with their nations' strategic importance in the broader geopolitical context surrounding the conflict.

The substantial coverage of Peter Dutton, Australia’s opposition leader, represents an unexpected finding. This anomaly may be attributed to his particular resonance with British media audiences rather than direct relevance to the conflict dynamics.

Brittney Griner’s appearance in the dataset, despite her non-political status, demonstrates how the war created ancillary diplomatic complications. As an American basketball player detained in Russia, her case became emblematic of the deteriorating US-Russia relations concurrent with the Ukraine conflict, justifying her inclusion in war-related coverage despite her civilian status.

Ukraine Administrative Regions mentioned in the Ukraine War news

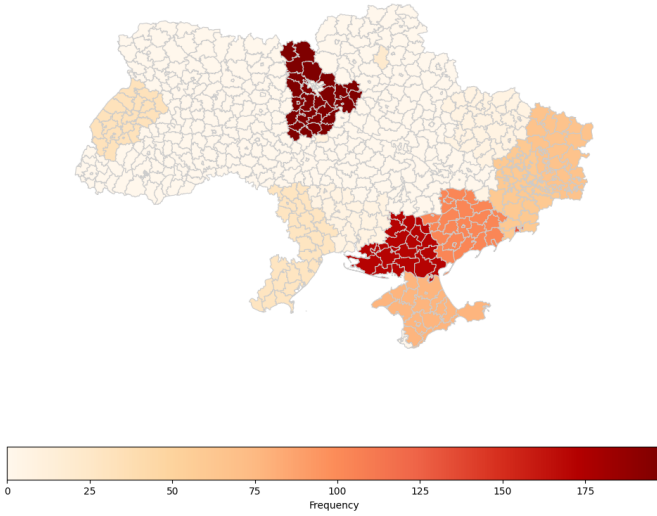


Fig. 7. Heatmap of top areas covered in Russo-Ukraine war news titles

3) *Heat Map Analysis of Regional Focus on Ukraine War Media Headline Coverage:* Subsequently, we constructed a geographical visualization to identify regions most frequently mentioned in Ukraine War news headlines. From the map analysis, we can identify several key regions that garnered significant media attention: Kyiv, Kherson, Zaporizhzhia, Donetsk, Luhansk, Crimea, Odesa, and Lviv. (See Figure 7 for heatmap, figure 8 for frequency count)

The prominence of Kyiv in media coverage is expected given its status as Ukraine’s capital, the location of the Chernobyl nuclear power plant, and the site of the Bucha massacre. These factors contributed to extensive media attention despite Kyiv no longer being on the frontline since the Russian military’s northern withdrawal in mid-2022.

Kherson Oblast and Zaporizhzhia Oblast represent significant conflict zones where Ukrainian forces initially experienced setbacks early in the war before launching a major counteroffensive in 2023. While Ukrainian forces achieved some territorial gains in Kherson, progress was limited in Zaporizhzhia Oblast, with both regions remaining active frontlines between the opposing forces.

Crimea and the Donbas region (encompassing Luhansk and Donetsk) have featured prominently in coverage due to their demographic composition, with significant Russian ethnic populations. Their attempts at independence from Ukraine precipitated the long-standing conflict between the two nations since 2014, justifying the media’s concentrated focus on these territories.

Lviv and Odesa garnered substantial media coverage due to their strategic significance—Lviv serving as a crucial logistical hub connecting Ukraine to Poland and NATO allies, while Odesa functions as Ukraine’s principal port. Both locations experienced repeated Russian aerial bombardments, thereby maintaining a consistent presence in international media reporting.

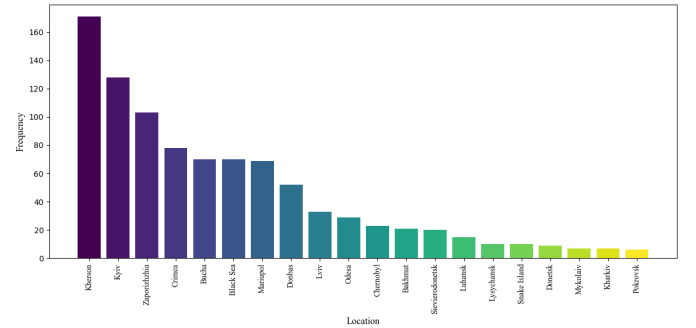


Fig. 8. Top locations in Ukraine covered in Russo-Ukraine war news titles

E. Interactive Map

We made an interactive visualization of the top named countries in the Russo-Ukraine war news titles. You can have a look at this link. You can also check a screenshot version of the interactive map, presented as Figure 9.

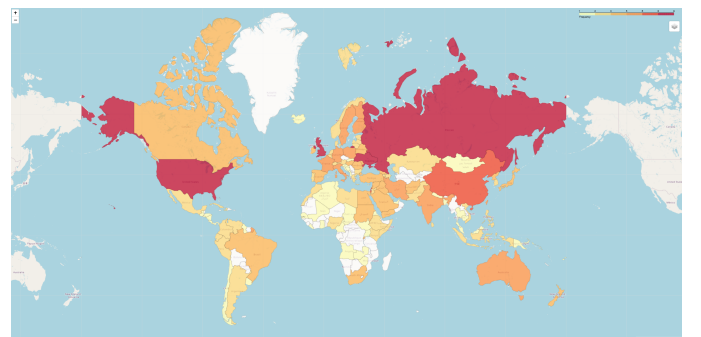


Fig. 9. Screenshot version of the interactive country-level heatmap in Russo-Ukraine war news titles (log level)

IV. CONCLUSION

We comprehensively examined the global news discourse surrounding the Russo-Ukrainian War by collecting and analyzing articles from BBC, Reuters, The Guardian, and The

New York Times. From the LDA topic modeling results, we extracted five primary topics: war actors, conflict events, international diplomatic efforts, economic impacts, and the Israel-Gaza situation. Semantic analysis of news headlines revealed predominantly positive sentiment throughout most of the period studied. Additionally, we created visualizations of the most frequently mentioned countries, world leaders, and conflict zones. Our analysis determined that beyond Russia and Ukraine, the United States, United Kingdom, China, and Israel received significant coverage in news headlines. Among world leaders, Putin, Biden, Johnson, and Zelensky were mentioned most frequently. The most referenced conflict areas were Kyiv, Kherson, Zaporizhzhia, and Donbas, which aligned with regions experiencing intense military activity.

V. AUTHORS & ACKNOWLEDGMENT

- **Zhian Li**: Web scrapping, LDA topic modeling, sentimental analysis
- **Haojian Li**: Web scrapping, visualization, interactive visualization

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